

Windsor App — Software Requirements Specification

Windsor is a web application for booking and managing event halls. It connects clients, service providers, administrators, and security staff on a single platform to reserve venues, hire complementary services, manage guest lists, and control access at events through QR codes.

Module Overview

Authentication and Access Control

This module manages user registration and login for clients and providers, using JWT tokens to keep sessions secure. It also restricts what each user can see and do based on their role (Administrator, Client, Provider, or Security Guard), and lets users recover their password through a temporary token sent by email.

Event Halls

This module lets the Administrator register event halls with their name, location, maximum capacity, and hourly cost, upload photos of each hall, and check availability by date and time range. It also keeps a booking schedule that prevents overlapping or duplicate reservations, and allows the Administrator to block dates or time slots for maintenance or private events.

Client Management

This module gives clients a profile where they can view and update their personal data, contact information, and notification preferences. Clients can also see a detailed history of their past, active, and cancelled events with invoices or summaries, and manage their guest list and hired provider services from a dedicated panel for each active event.

Providers and Services

This module allows providers to register and update the services they offer (catering, entertainment, music, decoration, etc.) along with descriptions and prices, and to attach one or more of these services to a hall reservation. Providers can view a calendar with the dates, times, and halls where they must deliver their services, and clients can rate and review a provider once an event has concluded.

Purchases and Payments

This module calculates the total cost of an event by adding the hall's hourly cost to the accumulated cost of the selected provider services. It generates and stores a purchase order linked to the client and the event, allows the Administrator or the payment gateway to update the payment status (Pending, Completed, Refunded), and produces a detailed digital receipt that the client can view or download.

Guests and QR Access Control

This module lets clients add guests one by one or through a bulk upload with contact details, and automatically generates a unique QR code for each guest and event. Security guards use the web application to scan and validate each QR code at the door; once a code is successfully scanned it is marked as "Used" so it cannot be reused, and clients can resend a guest's QR code individually if it gets lost.

Email Notifications

This module automatically emails clients to confirm their hall booking and hired services, sends each registered guest their QR-coded invitation, and notifies clients, providers, and guests by email whenever an event's date

changes or the event is cancelled. It also alerts the Administrator when a new provider requests registration or when an external integration fails.

Functional Requirements

ID	Requirement	Description
RF-AUTH-001	User registration	The system must allow clients and providers to register by entering their basic information and access credentials.
RF-AUTH-002	JWT authentication	The system must validate user credentials and generate a JWT token to manage secure sessions.
RF-AUTH-003	Role-based access control	The system must restrict views and actions according to the assigned role (Administrator, Client, Provider, Security Guard).
RF-AUTH-004	Password recovery	The system must allow users to recover access through a temporary token sent to their registered email.
RF-SAL-001	Event hall registration	The system must allow the Administrator to register halls, defining name, location, maximum capacity, and hourly cost.
RF-SAL-002	Hourly availability lookup	The system must allow checking a hall's availability on any date of the year, filtering by specific time ranges.
RF-SAL-003	Booking schedule	The system must record bookings by time range, preventing overlap or duplicate schedules for the same hall.
RF-SAL-004	Hall image upload	The system must allow uploading and storing photographs of the halls.
RF-SAL-005	Date blocking for maintenance	The system must allow the Administrator to disable halls or specific time ranges on given days for maintenance or private events.
RF-CLI-001	Client profile management	The system must allow the Client to view and update their personal data, contact information, and notification methods.
RF-CLI-002	Event history	The system must show the Client a detailed list of their past, active, and cancelled events with their respective invoices or summaries.
RF-CLI-003	Event management panel	The system must provide the Client with a panel to manage the guest list and the hired provider services for an active event.
RF-PROV-001	Provider service registration	The system must allow Providers to register and update their services (catering, entertainment, music, decoration) with descriptions and prices.
RF-PROV-002	Linking services to events	The system must allow attaching one or more provider services to an event hall booking.
RF-PROV-003	Provider schedule lookup	The system must offer Providers a calendar to view the dates, times, and halls where they must deliver their services.
RF-PROV-004	Provider rating	The system must allow the Client to rate and leave a review of the services provided by the Provider once the event has concluded.
RF-COMP-001	Total purchase cost calculation	The system must calculate the total amount by adding the hourly hall cost and the accumulated cost of the selected provider services.
RF-COMP-002	Purchase order registration	The system must generate and store the purchase order in pending or paid status, associating it with the client and the event.
RF-COMP-003	Transaction status change	The system must allow the Administrator or the payment gateway to update the payment status (Pending, Completed, Refunded).
RF-COMP-004	Purchase receipt generation	The system must generate a detailed summary of the purchase in digital format for the client to view or download.
RF-INV-001	Guest list upload	The system must allow the Client to enter guests individually or perform a bulk upload with contact details.
RF-INV-002	Unique QR code generation	The system must generate a unique hash and represent it graphically as an exclusive QR code for each guest and registered event.

ID	Requirement	Description
RF-INV-003	QR scanning and validation	The system must allow the Security Guard to validate the authenticity of the QR code presented by the guest through the web application.
RF-INV-004	Real-time access control	The system must mark the QR code as "Used" after the first successful scan to prevent the same entry from being reused.
RF-INV-005	QR code resend	The system must allow the Client to individually resend the QR-coded ticket if a guest loses it.
RF-MAIL-001	Purchase confirmation notification	The system must send an automatic email to the Client confirming the hall booking and the hired services.
RF-MAIL-002	Bulk QR access delivery	The system must automatically send invitations with the packaged QR code to the email address of each registered guest.
RF-MAIL-003	Schedule change notification	The system must notify Clients, Providers, and Guests by email of any change in date or cancellation of the event.
RF-MAIL-004	Administration alerts	The system must notify the Administrator when a new Provider requests registration or when a failure occurs in external integrations.

Non-Functional Requirements

ID	Requirement	Description
RNF-SEG-001	Reverse proxy concealment	The NestJS monolithic backend must operate behind a Proxy server (Nginx) without publicly exposing its private IP or ports.
RNF-SEG-002	Database isolation	The database must be hosted on a physically independent server with no public access, accepting connections only from the backend's IP.
RNF-PER-002	API response time	Requests processed by the NestJS routes must respond with an average time under 300 ms under normal traffic conditions.
RNF-USAB-001	Responsive interface	The Angular web application must adapt its layout to desktop browsers and mobile devices to facilitate QR reading at the door.
RNF-DISP-001	Service availability	The system must maintain 99.5% operational availability throughout the year to guarantee real-time queries and validations.
RNF-COMP-001	Scanning compatibility	The QR validation module must be compatible with the cameras of mobile devices running modern browsers (Chrome, Safari).

Architecture

Windsor: Party Hall Management System ARchitecture

